

Algebraic Geometry Cheat Sheet

Affine Space

Let K be an algebraically closed field. The **affine n -space** is $\mathbb{A}^n = K^n$ as a set. Write $K[x_1, \dots, x_n]$ for the polynomial ring and $f(x)$ for evaluation at a point $x \in \mathbb{A}^n$.

Ideal of a Variety

For a set $X \subseteq \mathbb{A}^n$ the **ideal of a variety** is $I(X) := \{f \in K[x_1, \dots, x_n] : f(x) = 0 \forall x \in X\}$.

The **coordinate ring** is $A(X) = K[x_1, \dots, x_n]/I(X)$.

- $X \subseteq Y \implies I(X) \supseteq I(Y)$
- $I(X \cup Y) = I(X) \cap I(Y)$
- $I(X \cap Y) = \sqrt{I(X) + I(Y)}$
- $I(V(I)) = \sqrt{I}$

Affine Varieties

For $S \subseteq K[x_1, \dots, x_n]$ the **zero locus** is $V(S) := \{x \in \mathbb{A}^n : f(x) = 0 \forall f \in S\}$. Subsets of this form are **affine varieties**.

- $S_1 \subseteq S_2 \implies V(S_1) \supseteq V(S_2)$
- $V(S_1) \cup V(S_2) = V(S_1 S_2)$
- $\bigcap_i V(S_i) = V(\sum_i S_i)$

When S is an ideal:

- $V(I) = V(\sqrt{I})$
- $V(I_1) \cup V(I_2) = V(I_1 I_2) = V(I_1 \cap I_2)$
- $V(I_1) \cap V(I_2) = V(I_1 + I_2)$

Nullstellensatz

Nullstellensatz:

- Weak:** Every proper ideal $I \subsetneq K[x_1, \dots, x_n]$ has a zero ($V(I) \neq \emptyset$)
- Hilbert:** Order-reversing bijection between radical ideals and affine varieties via $I \leftrightarrow V(I)$. This is an **adjoint functor**.
- Relative:** $f \in \sqrt{I}$ iff f vanishes on $V(I)$.

Coordinate Rings and Relative Theory

Coordinate ring: for affine $X \subset \mathbb{A}^n$, $A(X) = K[x_1, \dots, x_n]/I(X)$, and regular functions on X are exactly elements of $A(X)$.

For affine Y and $X \subset Y$ affine:

- Relative zero loci and ideals are defined in $A(Y)$.
- Relative Nullstellensatz gives an inclusion-reversing bijection between affine subvarieties of Y and radical ideals of $A(Y)$.
- $A(X) \cong A(Y)/I_Y(X)$.

Zariski Topology

For affine X , the closed sets are exactly affine subvarieties $V(S)$ with $S \subset A(X)$.

A **distinguished open** is $D(f) := X \setminus V(f)$ for $f \in A(X)$.

- $D(f) \cap D(g) = D(fg)$
- $D(1) = X$, $D(0) = \emptyset$

Key facts:

- Zariski topology on X agrees with the subspace Zariski topology from any affine ambient variety.
- In $\mathbb{A}^1$, closed sets are finite sets or all of $\mathbb{A}^1$.
- Injective maps $\mathbb{A}^1 \rightarrow \mathbb{A}^1$ are automatically Zariski-continuous.
- Zariski topology on $X \times Y$ is usually not the product topology.
- $V(I(T)) = \overline{T}$ for every $T \subset \mathbb{A}^n$.
- Every Noetherian topological space is compact.

Irreducibility and Connectedness

For a topological space X :

- Reducible:** $X = X_1 \cup X_2$ with closed $X_i \subsetneq X$.
- Disconnected:** $X = X_1 \cup X_2$, $X_1 \cap X_2 = \emptyset$, closed proper X_i .
- Noetherian:** no infinite strictly descending chain of closed sets.

For affine varieties:

- If X is disconnected, then $A(X) \cong A(X_1) \times A(X_2)$.
- Non-empty X is **irreducible** iff $A(X)$ is an integral domain.
- Irreducible affine subvarieties of Y correspond to prime ideals of $A(Y)$.
- Every affine variety is Noetherian and has a finite irreducible decomposition, unique up to order.
- Irreducible components correspond to minimal prime ideals.
- In irreducible X , any two non-empty open subsets intersect and every non-empty open subset is dense.
- Continuous images preserve connectedness and irreducibility.
- If X, Y are irreducible affine varieties, then $X \times Y$ is irreducible.
- Ideal quotients encode differences: $I(Y_1 \setminus Y_2) = I(Y_1) : I(Y_2)$ and $V(J_1) \setminus V(J_2) = V(J_1 : J_2)$.

Dimension of a Variety

The **dimension** of non-empty X is the maximal length of chains $\emptyset \neq Y_0 \subsetneq \dots \subsetneq Y_n \subset X$ with irreducible closed Y_i .

For irreducible closed $Y \subset X$, the **codimension** in X is the maximal chain length from Y to X .

For affine varieties:

- $\dim X$ equals the Krull dimension of $A(X)$.
- $\text{codim}_X(Y) = \text{codim}(I(Y) \subset A(X))$.
- Dimensions and codimensions are finite.
- $\dim(X \times Y) = \dim X + \dim Y$; in particular $\dim \mathbb{A}^n = n$.
- If $Y \subset X$ is irreducible, then $\dim X = \dim Y + \text{codim}_X(Y)$.
- If $0 \neq f \in A(X)$, each irreducible component of $V(f)$ has codimension 1.
- $\dim X = \max_i \dim X_i$ over irreducible components.
- $\dim X = \sup\{\text{codim}_X\{a\} : a \in X\}$.
- $\text{codim}_X\{a\} = \max\{\dim X_i : a \in X_i\}$.
- If $A \subset X$, then $\dim A \leq \dim X$.
- For an open cover $X = \bigcup_i U_i$, $\dim X = \sup_i \dim U_i$.
- If X is irreducible affine and $U \subset X$ is non-empty open, then $\dim U = \dim X$.

Hypersurfaces and UFD

For a Noetherian integral domain R , the following are equivalent:

- Every prime ideal of codimension 1 is principal.
- R is a unique factorization domain.

For hypersurfaces in affine space:

- In $K[x_1, \dots, x_n]$, every affine hypersurface has ideal $\langle f \rangle$.
- If the hypersurface is irreducible, then f is irreducible.
- If $I(X) = \langle f \rangle$, then **degree**(X) := $\deg f$.
- Names by degree: linear (1), quadric (2), cubic (3).

Pure-dimensional terminology:

- X is pure-dimensional of dimension n if all irreducible components have dimension n .
- Curve: pure dimension 1. Surface: pure dimension 2.
- Hypersurface in pure-dimensional Y : pure dimension $\dim Y - 1$.

Counterexample:

- $R = K[x_1, x_2, x_3, x_4]/\langle x_1 x_4 - x_2 x_3 \rangle$ is not a UFD.
- It has a codimension-1 prime ideal that is not principal.

Regular Functions

Let X be affine and $U \subset X$ open. A map $\varphi : U \rightarrow K$ is **regular** if every $a \in U$ has a neighborhood U_a with $\varphi = \frac{g}{f}$ on U_a , for $f, g \in A(X)$ and $f \neq 0$ on U_a .

- The regular functions on U form a K -algebra $\mathcal{O}_X(U)$.
- For $\varphi \in \mathcal{O}_X(U)$, its zero locus $V(\varphi)$ is closed in U .
- Identity theorem: on irreducible X , if two regular functions on an open set agree on a non-empty open subset, they agree everywhere.

Distinguished Open Subsets and Localization

For $f \in A(X)$, set $D(f) := X \setminus V(f)$.

- $D(f) \cap D(g) = D(fg)$.
- Every open subset is a finite union of distinguished opens.
- $\mathcal{O}_X(D(f)) = \left\{ \frac{g}{f^n} : g \in A(X), n \in \mathbb{N} \right\} \cong A(X)_f$.
- Hence $\mathcal{O}_X(X) = A(X)$.

If $A(X)$ is a UFD and $U = X \setminus Y$, then $\mathcal{O}_X(U) = A(X)$ iff $\text{codim } Y \geq 2$.

Sheaves and Stalks of Regular Functions

A **presheaf** on X consists of:

- a ring $\mathcal{F}(U)$ for each open $U \subset X$;
- restriction homomorphisms $\rho_{V,U} : \mathcal{F}(V) \rightarrow \mathcal{F}(U)$ for $U \subset V$, with identity and composition compatibility.

A **sheaf** is a presheaf with unique gluing of compatible local sections.

- $\mathcal{O}_X$ is a sheaf of K -algebras on X .
- For $a \in X$, the **stalk** $\mathcal{O}_{X,a}$ is the ring of germs of regular functions at a .
- Algebraically, $\mathcal{O}_{X,a} \cong A(X)_{I(a)} = \left\{ \frac{g}{f} : f(a) \neq 0 \right\}$.
- $\mathcal{O}_{X,a}$ is local with maximal ideal $I_a = \{(U, \varphi) : \varphi(a) = 0\}$.

Sheaf Operations

For a sheaf $\mathcal{F}$ on X :

- Restriction to $U \subset X$: $\mathcal{F}|_U(V) = \mathcal{F}(V)$ for $V \subset U$ open.
- Stalk at $a \in X$: germs of sections near a .
- For an irreducible closed $Y \subset X$, the stalk at Y corresponds to functions near dense open subsets of Y .

For affine X : $\mathcal{O}_{X,a} \cong A(X)_{I(a)}$ and $\mathcal{O}_{X,Y} \cong A(X)_{I(Y)}$ for irreducible closed Y .

Morphisms of Ringed Spaces

A **ringed space** is a topological space X with a structure sheaf $\mathcal{O}_X$. An affine variety is viewed with its sheaf of regular functions.

For a map $f : X \rightarrow Y$, pull-back is $f^* \varphi := \varphi \circ f$. The map f is a **morphism** if:

- f is continuous;
- for every open $U \subset Y$ and $\varphi \in \mathcal{O}_Y(U)$, $f^* \varphi \in \mathcal{O}_X(f^{-1}(U))$.

A bijective morphism need not be an isomorphism.

Affine Morphisms

If $U \subset X$ is open and $Y \subset \mathbb{A}^n$ affine, then morphisms $f : U \rightarrow Y$ are exactly maps $f = (\varphi_1, \dots, \varphi_n)$ with $\varphi_i \in \mathcal{O}_X(U)$ and image in Y .

For affine varieties X, Y : $\text{Hom}(X, Y) \cong \text{Hom}_{K\text{-alg}}(A(Y), A(X))$, $f \mapsto f^*$. Hence isomorphisms of affine varieties correspond to isomorphisms of coordinate rings.

Finite type reduced K -algebras and affine varieties correspond up to isomorphism.

Product Varieties

Product varieties satisfy the universal property: for morphisms $f_X : Z \rightarrow X$ and $f_Y : Z \rightarrow Y$ there is a unique $f : Z \rightarrow X \times Y$ with $\pi_X \circ f = f_X$ and $\pi_Y \circ f = f_Y$.

So giving a morphism to $X \times Y$ is equivalent to giving one to each factor.

On coordinate rings, $A(X \times Y) \cong A(X) \otimes_K A(Y)$.

Distinguished Opens as Affine Varieties

For affine X and $f \in A(X)$, the distinguished open $D(f) = X \setminus V(f)$ is affine, with coordinate ring $A(D(f)) \cong A(X)_f$.

This realizes localization geometrically.

$\mathbb{A}^2 \setminus \{0\}$ is not affine, although it is covered by the affine opens $D(x_1)$ and $D(x_2)$.

Locality and Reconstruction of Morphisms

Morphisms are stable under composition and restriction. They satisfy a gluing property: if a map is a morphism on an open cover, then it is a morphism globally.

Further structure facts:

- Affine varieties correspond to finitely generated reduced K -algebras (up to isomorphism).
- Affine variety is used up to isomorphism of ringed spaces.
- For affine X and $f \in A(X)$, $D(f)$ is affine with coordinate ring $A(X)_f$.
- $\mathbb{A}^2 \setminus \{0\}$ is not affine, though covered by affine opens.

Prevarieties and Gluing

A **prevariety** is a ringed space with a finite open cover by affine varieties. Affine varieties and their open subsets are prevarieties.

Gluing principle:

- Glue prevarieties X_i along isomorphic open subsets $U_{i,j}$.
- Compatibility cocycle conditions give a well-defined glued prevariety.

Open and Closed Subprevarieties

For a prevariety X :

- Open subsets are open subprevarieties with restricted sheaf.
- Closed subsets carry a natural sheaf of locally restricted regular functions, hence closed subprevarieties.
- Inclusion $Y \hookrightarrow X$ of a closed subprevariety is a morphism.
- If $f : Z \rightarrow X$ and $f(Z) \subset Y$, then f factors as a morphism $Z \rightarrow Y$.
- Inverse images of open/closed subprevarieties are open/closed.

Product Prevarieties

Product $X \times Y$ of prevarieties is characterized by the universal property: for morphisms $f_X : Z \rightarrow X$, $f_Y : Z \rightarrow Y$, there is a unique $f : Z \rightarrow X \times Y$ with $\pi_X \circ f = f_X$ and $\pi_Y \circ f = f_Y$.

Existence: glue affine products of affine open patches. Uniqueness: unique up to unique isomorphism.

If X', Y' are closed subprevarieties, then the two natural prevariety structures on $X' \times Y'$ agree.

Varieties and Diagonal Criterion

A prevariety X is a **variety** (separated) iff the diagonal $\Delta_X = \{(x, x) : x \in X\}$ is closed in $X \times X$.

Key results:

- Affine varieties are varieties.
- Open and closed subprevarieties of varieties are varieties.
- For pure-dimensional varieties, curves and surfaces are dimensions 1 and 2.
- For morphisms to a variety Y , graph Γ_f is closed and equalizer $\{x : f(x) = g(x)\}$ is closed.